



An individual-specific gait pattern prediction model based on generalized regression neural networks



Trieu Phat Luu^{*}, K.H. Low, Xingda Qu, H.B. Lim, K.H. Hoon

School of Mechanical and Aerospace Engineering, Nanyang Technological University, Singapore

ARTICLE INFO

Article history:

Received 23 May 2012

Received in revised form 19 August 2013

Accepted 25 August 2013

Keywords:

Gait pattern planning

Robotic gait rehabilitation

Lower limb angular kinematics

ABSTRACT

Robotics is gaining its popularity in gait rehabilitation. Gait pattern planning is important to ensure that the gait patterns induced by robotic systems are tailored to each individual and varying walking speed. Most research groups planned gait patterns for their robotics systems based on Clinical Gait Analysis (CGA) data. The major problem with the method using the CGA data is that it cannot accommodate inter-subject differences. In addition, CGA data is limited to only one walking speed as per the published data. The objective of this work was to develop an individual-specific gait pattern prediction model for gait pattern planning in the robotic gait rehabilitation systems. The waveforms of lower limb joint angles in the sagittal plane during walking were obtained with a motion capture system. Each waveform was represented and reconstructed by a Fourier coefficient vector which consisted of eleven elements. Generalized regression neural networks (GRNNs) were designed to predict Fourier coefficient vectors from given gait parameters and lower limb anthropometric data. The generated waveforms from the predicted Fourier coefficient vectors were compared to the actual waveforms and CGA waveforms by using the assessment parameters of correlation coefficients, mean absolute deviation (MAD) and threshold absolute deviation (TAD). The results showed that lower limb joint angle waveforms generated by the gait pattern prediction model were closer to the actual waveforms compared to the CGA waveforms.

© 2013 Elsevier B.V. All rights reserved.

1. Introduction

Robotic assisted gait rehabilitation has gained extensive attentions from researchers and clinical practitioners in recent years [1]. For example, several robotic devices, such as Lokomat [2], WalkTrainer [3], HapticWalker [4], ALEX [5] and KineAssist [6], have been developed to enhance motor function and ambulation in individuals with spinal cord (SCI) and stroke injuries. Specificity is one of the major principles of gait rehabilitation [7,8], which suggests that it is crucial to induce a gait pattern that resembles natural gait pattern of the specific targeted subject during gait rehabilitation.

Lower limb joint kinematics was well documented (e.g. CGA database [9]) and can be easily used in robotic gait rehabilitation systems. Previous studies mainly used CGA data as the reference in their gait pattern planning [3,5,10–12]. However, gait data in the CGA database were only recorded from limited population. Therefore, the CGA data cannot well account for inter-individual

differences. Furthermore, CGA data is limited to one recorded walking speed for each individual, and thus cannot allow the changing of walking speed in gait pattern planning. The influence of walking speed on kinematic and kinetic gait parameters has been presented in many studies [13,14]. Schwartz et al. [13] found that local extrema of walking gait pattern varied with walking speed in both linear and nonlinear ways. However, a model that generates gait pattern given walking speed as one of input parameters has not been presented.

The objective of this work is to develop an individual-specific gait pattern prediction model that is able to generate walking gait patterns based on gait parameters (stride length, cadence) and anthropometric data of targeted subjects. We hypothesized that the proposed model would be able to generate a gait pattern better than the CGA approach in terms of resembling the actual gait pattern of the targeted subject.

2. Methods

2.1. Experimental setup

Seventy healthy adults with age ranging from 18 to 33 (mean 23.6 ± 2.2 years), height 156–192 (mean 171.1 ± 6.2 cm) and body mass 49–120 (mean 64.3 ± 11.5 kg), were recruited from Nanyang

^{*} Corresponding author at: School of Mechanical and Aerospace Engineering, Nanyang Technological University, Blk N3, North Spine, Nanyang Avenue, Singapore 639798, Singapore. Tel.: +65 67905568.

E-mail addresses: luut0004@e.ntu.edu.sg, phatct2207@yahoo.com (T.P. Luu).

Technological University (NTU) in Singapore. These subjects have neither neurological injury nor gait disorder. Each subject was instructed to walk ten times at their self-paced slow and normal walking speeds on a twelve-meter walkway. The gait data were collected using an eight-camera motion analysis system (Eagle, Motion Capture Analysis Corp) and the placement of fifteen reflective markers is the same as that in [15]. Anthropometric data of each subject including ASIS breadth, thigh length, calf length and foot length were also measured.

Positions of the lower limb joint centers were estimated from the reflective markers and recorded anthropometric data. Subsequently, a segment reference frame in each lower limb segment was defined by the positions of joint centers and reflective markers. Joint angles were defined as a rotation of the distal segment frame relative to the proximal segment frame [15]. In this work, the angular kinematics in sagittal plane (flexion-extension) was used to verify the gait pattern prediction model.

2.2. Methodology of gait pattern prediction

The gait pattern prediction model is illustrated in Fig. 1. Fourier transform is a traditional spectral analysis method to describe any periodic signal in its harmonic components [16]. Since walking is periodic and the power for walking is supplied rhythmically with temporal consistency [9], Fourier transform is often used to describe the frequency content of gait [9,17]. In this study, each joint angle waveform was analyzed in frequency domain and decomposed into one Fourier coefficient vector. A database that consists of gait parameters, anthropometric data and Fourier coefficient vectors was constructed from the experimental data. Subsequently, an individual-specific gait pattern prediction model was designed based on this database. Non-linear mapping functions between the input vectors (gait parameters and anthropometric data) and Fourier coefficient output vectors were determined using GRNNs. With these mapping functions, Fourier coefficient vectors were predictable and inverse Fourier transform was then applied to generate joint kinematics for the subject.

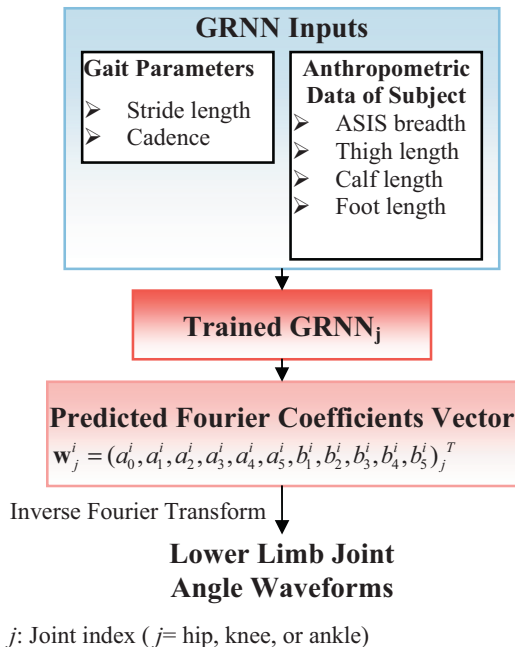


Fig. 1. Individual-specific gait pattern prediction model for robotic assisted gait rehabilitation.

2.3. Extraction of Fourier coefficient vectors

Prior to the extraction of Fourier coefficients of the joint angle waveforms, the frequency content of gait data must be determined. Power spectral density analysis was conducted on the collected experimental gait data and the average values were calculated. The results revealed that over 96.2% of the power of the hip, knee, and ankle joint angle waveforms are contained below 6 Hz. Thus, a 6 Hz low-pass filter was applied to reduce noise data of these signals. The result is agreed with the previous studies [9,17], which suggested that the cut-off frequency for waking data should be set at about 6 Hz.

Each lower limb joint angle waveform in one gait cycle can be decomposed to sinusoidal components as follows:

$$f(t_n) = \frac{1}{2}a_0 + \sum_{p=1}^{N/2} \left[a_p \cos\left(\frac{2\pi pn}{N}\right) + b_p \sin\left(\frac{2\pi pn}{N}\right) \right], \quad n = 0, 1, \dots, N \quad (1)$$

where a_p and b_p are the Fourier coefficients, and N is the samples taken for one cycle.

Fourier coefficients can be computed with the following formulas:

$$a_p = \frac{2}{N} \sum_{n=1}^N f(t_n) \cos\left(\frac{2\pi pn}{N}\right), \quad p = 0, 1, \dots, \frac{N}{2} - 1 \quad (2)$$

$$a_{N/2} = \frac{1}{N} \sum_{n=1}^N f(t_n) \cos(\pi n)$$

$$b_p = \frac{2}{N} \sum_{n=1}^N f(t_n) \sin\left(\frac{2\pi pn}{N}\right), \quad p = 0, 1, \dots, \frac{N}{2} \quad (3)$$

Since the sinusoidal components above 6 Hz are insignificant, the lower limb joint angles for any sample point, n , in one gait cycle can be represented by:

$$f(t_n) = \frac{1}{2}a_0 + \sum_{p=1}^5 \left[a_p \cos\left(\frac{2\pi pn}{N}\right) + b_p \sin\left(\frac{2\pi pn}{N}\right) \right] \quad (4)$$

Using the cut-off frequency as 6 Hz, Fourier coefficients vector is formed as:

$$u_j^i = (a_0^i, a_1^i, a_2^i, a_3^i, a_4^i, a_5^i, b_1^i, b_2^i, b_3^i, b_4^i, b_5^i)^T \quad (5)$$

where $(i = 1, 2, \dots, 700)$ is the walking trial number and j is the joint index ($j = \text{hip, knee, or ankle}$).

2.4. Design of generalized regression neural networks (GRNNs)

Neural network can be viewed as a highly flexible and nonlinear regression model and it has been a prevalent analysis tool used in gait studies [16]. In this study, GRNNs were applied to predict the Fourier coefficient vectors when the gait parameters and anthropometric data of the subjects are provided. The GRNNs in the present study were designed and verified using Matlab Neural Network Toolbox [18].

2.4.1. Architecture design of GRNNs

Based on the existing literature [19–23], stride length, cadence, ASIS breadth, thigh length, shank length, and foot length were found to be the key factors influencing the lower limb joint angle waveforms. Gait parameters were calculated from motion capture experimental data, while anthropometric data were measured from participants.

GRNN is often used as a powerful regression tool because it has a high training speed and a strong non-linear mapping capability

[24]. The network consists of an input layer, a radial basis layer and a linear output layer. Each neuron in the hidden layer has a radial basis function to perform the nonlinear transformation from the input space to the hidden space. The Gaussian function is the common choice for the radial basis neurons and it is given by:

$$g(d) = \exp\left(-\frac{d^2}{2 \cdot \sigma^2}\right) \quad (6)$$

where d is the distance between two vectors and σ is the spread of Gaussian function or also referred as smoothing factor.

In this study, seventy healthy adults were instructed to walk five times at self-paced slow and normal walking speeds, respectively. Experimental data of all 700 walking trials were divided into a training set (600 walking trials) and a test set (100 walking trials). The subjects in test set with ASIS breadth ranging from 22.4 to 28.4 cm, thigh length 36.3–50 cm, shank length 39–43 cm, and foot length 22.6–26.4 cm, were randomly selected. The input vector of the GRNN ($\mathbf{p}_j^i$) consists of gait parameters and anthropometric data, while the output vector ($\mathbf{q}_j^i$) is a Fourier coefficient vector.

2.4.2. Smoothing factor optimization

The smoothing factor plays an important role in function approximation [24,25]. A smaller smoothing factor will lead to a 'steeper' radial basis function, so that the neuron with the weight vector closest to the input vector will provide much larger output than other neurons. With an increase in the smoothing factor, the radial basis function's slope becomes smoother and more neurons can respond to an input vector. However, a larger smoothing factor will also make the network function less selective. Therefore, there is a need to determine the optimal smoothing factors for GRNNs. In this study, the optimal smoothing factors were determined by comparing the distances between the prediction and the actual Fourier coefficient vectors in the training set using cross validation. The obtained optimal smoothing factors for GRNN_{hip}, GRNN_{knee}, and GRNN_{ankle} are 25.6, 30.1, and 35.6, respectively.

2.5. Assessment parameters for gait pattern prediction model

GRNNs were implemented to predict the Fourier coefficient output vectors $\hat{\mathbf{w}}_j^i$ for the test set (100 walks). Subsequently, the predicted Fourier coefficient vectors $\hat{\mathbf{w}}_j^i$ were used to reconstruct the lower limb joint angle waveforms by applying the Inverse Fourier Transform (Eq. (4)). In order to assess the quality of gait pattern prediction, comparison was carried out to examine the difference between the predicted angles $\hat{\theta}$ and the actual angles θ . The quality of gait pattern prediction model was evaluated by using three assessment parameters: correlation coefficient ρ , mean absolute deviation (MAD) and threshold absolute deviation (TAD). The definitions of MAD and TAD are given as follows:

$$MAD_j = \sum_{n=0}^{N-1} \frac{|\hat{\theta}_{j,n} - \theta_{j,n}|}{N} \quad (7)$$

where N is the number of samples in one gait cycle collected from the motion capture experiment, j is the joint index, θ_j and $\hat{\theta}_j$ are the actual and predicted angles of joint j .

$$TAD_j = \frac{100}{N} \sum_{n=0}^{N-1} T_{j,n} \quad (8)$$

$$T_{j,n} = \begin{cases} 1, & \text{when } |\hat{\theta}_{j,n} - \theta_{j,n}| > \gamma \\ 0, & \text{otherwise} \end{cases} \quad (9)$$

TAD determines the percentage of points which exceed a pre-specified threshold around the actual waveform. The pre-specified threshold used here is 5° , which is consistent with a previous study [26].

CGA database was often used in gait pattern planning [27]. Normative gait database from Winter [9] is a reliable and popular resource of CGA. To evaluate the gait pattern prediction model, the predicted kinematics was compared with CGA data using the assessment parameters. The differences of assessment parameters were also evaluated. In particular, t-tests were used to determine if the differences were statistically significant. The level of significance was set at 0.05.

3. Results

3.1. Verification of representing lower limb joint angle waveforms by Fourier coefficient vectors

The reconstructed and actual joint angle waveforms associated with the 700 walking trials were generated for comparison. Assessment parameters were calculated to measure the differences between the reconstructed and the actual waveforms (Table 1). The results show very high correlations (0.99–1.00), low MAD ($\leq 0.8^\circ$), and small TAD ($\leq 0.1\%$).

3.2. Validation of gait pattern prediction model

The actual and predicted joint angle waveforms associated with 100 walking trials are shown in Fig. 2, which indicates that the predicted waveforms have similar patterns as the actual waveforms. In general, the average predicted waveforms fall within one standard deviation of actual waveforms. Variations from predicted waveforms exceed one standard deviation of actual waveforms at some instant but the values are less than 4.4° .

Assessment parameters (ρ , MAD, and TAD) were calculated to verify the quality of the gait pattern prediction model and to compare the model with the CGA approach [9]. Mean and standard deviation of assessment parameters from 100 walks in test set were calculated and presented in Table 2.

MAD values highlighted that the average deviation of predicted waveforms from the actual ones ranged from 3.73 to 5.41° . For CGA, the average value of MAD is as high as 9.28° , while the lowest is 6.03° .

When using gait pattern prediction model, the maximum TAD (i.e. 44.94%) and minimum TAD (i.e. 24.64%) percentages were found for the knee and ankle joint, respectively. These values are much lower compared to those from CGA, which ranged from 47.35% to 64.62%.

The values of MAD and TAD calculated using the present model were statistically smaller compared to using CGA data ($p < 0.001$). The confidence intervals of MAD differences for hip, knee, and ankle waveforms are (3.94 ± 0.38) , (3.87 ± 0.96) , and (2.46 ± 0.58) degrees, respectively. Similarly, the confidence intervals of TAD differences are (34.98 ± 5.12) , (17.35 ± 7.05) and (22.71 ± 6.88) .

Table 1
Mean (Std) of assessment parameters from 700 walks.

	ρ	MAD ($^\circ$)	TAD (%)
Hip	1.00 (0.00)	0.0 (0.0)	0.0 (0.0)
Knee	0.99 (0.02)	0.5 (0.2)	0.0 (0.3)
Ankle	0.98 (0.03)	0.8 (0.3)	0.1 (0.7)

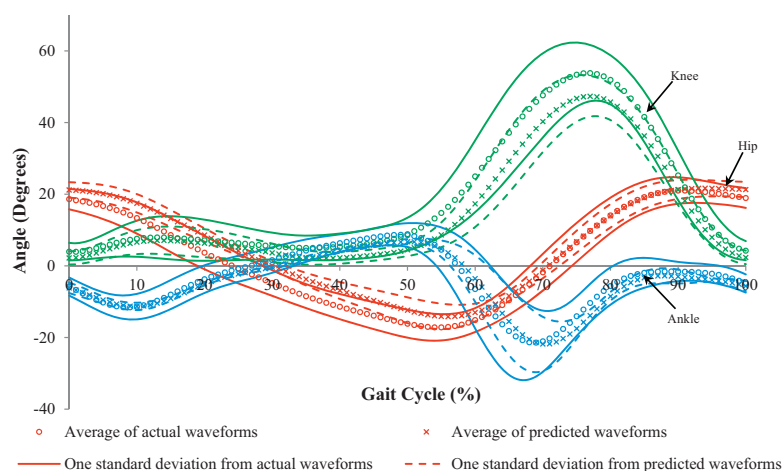


Fig. 2. Actual lower limb joint angle waveforms and predicted waveforms from gait pattern prediction model associated with 100 walks.

Table 2

Mean (Std) of assessment parameters from 100 walks in test set.

	ρ		MAD ($^{\circ}$)		TAD (%)	
	Gait pattern prediction model	CGA from Winter [17]	Gait pattern prediction model	CGA from Winter [17]	Gait pattern prediction model	CGA from Winter [17]
Hip	0.98 (0.03)	0.87 (0.07)	3.73 (1.64)	7.66 (1.78)	29.64 (23.13)	64.62 (14.73)
Knee	0.97 (0.04)	0.85 (0.11)	5.41 (2.01)	9.28 (3.07)	44.94 (23.19)	62.29 (15.21)
Ankle	0.92 (0.07)	0.58 (0.34)	3.58 (1.44)	6.03 (1.68)	24.64 (20.47)	47.35 (16.86)

3.3. Response of gait pattern prediction model with varying walking speed

The responses of gait pattern prediction model when different walking speeds applied are shown in Fig. 3. The predicted hip and knee joint angle waveforms are updated in both spatio- and temporal-space when walking speed is changed from 0.11 to 0.25 m/s. For example, $Peak_a$ values of hip and knee are updated from 21.5° to 23.5° and 47.9° to 48.6° , respectively; $Peak_b$ values are updated from -14.1° to -16.1° and 6.7° to 9.6° ; and the time to complete one gait cycle is reduced from 94.8 to 57.2 s.

3.4. Implementation of gait pattern prediction model in robotic gait rehabilitation

The present model has been applied to generate gait patterns for an over-ground gait trainer developed in NTU to study its

applicability in robotic gait rehabilitation. The system was named as *NaTure-gaits* and details can be found in [28]. A 64 year old male subject with incomplete spinal cord injury (iSCI) has been recruited for clinical trial carried out in the Tan Tock Seng Hospital in Singapore. Basic information of the subject (age, gender, height, weight) was recorded and anthropometric data were measured. When walking speed was determined, gait parameters were estimated [29] and the present model was applied to generate gait patterns. The average slow and comfortable walking speeds in healthy male adults calculated from 700 walks are 0.55 and 1.30 m/s. The gait patterns were played at a slower cadence on *NaTure-gaits* because comfortable walking speed of subjects with severe gait deficit is less than 0.4 m/s [30]. The results were positive since the subject did not experience unnatural and great restrictive motion during the trial. The comfort level is acceptable based on feedbacks from the subject.

4. Discussion

An individual-specific gait pattern prediction model was presented in this work. The model involves two major components. First, FFT was used to simplify the joint angle waveforms. The simplification is able to facilitate the analysis process, whereby each waveform is significantly reduced to eleven Fourier coefficients. The assessment parameters in Table 1 show very high correlations (0.99–1.00), low MAD ($\leq 0.8^{\circ}$), and small TAD ($\leq 0.1\%$) between the reconstructed and the actual waveforms. The results demonstrate that each joint angle waveform can be represented by a Fourier coefficient vector.

The second component of the present model is the nonlinear mapping function, which predicts Fourier coefficient vectors from given gait parameters and anthropometric data of an individual. The lower limb angular kinematics can be generated from the predicted Fourier coefficient vector by using the inverse Fourier transform (Eq. (4)).

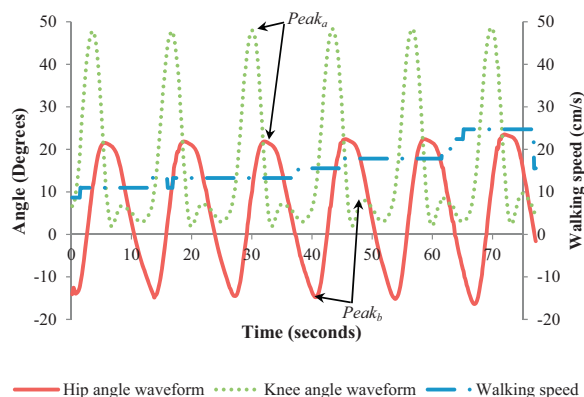


Fig. 3. Joint kinematics of hip and knee joints in sagittal plane generated by gait pattern prediction model with varying walking speed.

The accuracy of the model was assessed by using the correlation coefficient, *MAD* and *TAD*. These parameters showed that the present model was able to generate gait patterns closer to the actual patterns compared to the gait patterns selected from CGA. These findings support the hypothesis that the proposed model is able to generate a gait pattern better than the CGA approach in terms of resembling the actual gait pattern of the targeted subject.

Although the present model was shown to generate a better gait pattern compared to CGA method, the predicted waveforms still exceed one standard deviation of actual waveforms at some instant (Fig. 2). The variation may cause improper foot contact during stance phase of human walking. For example, heel trajectories in stance phase calculated using lower limb kinematics model with variation of lower limb joint angles may travel below the ground level or “float” above it.

Different walking speeds used in the robotic gait rehabilitation systems may be required at different stages of the training process. For example, slow walking speeds are usually used for training at the beginning stage. As the condition is improved, faster walking speeds are applied to provide a better therapeutic outcome. Walking speed is one of the key factors that influence the walking gait pattern [21,22]. Therefore, if different walking speeds are required for the training purpose, suitable gait patterns must be planned for the robotic systems. Fig. 3 shows that the present model is able to update gait patterns in both spatio- and temporal-space when walking speed is changed. The plots also illustrate that the peaks of hip and knee joint angle waveforms are increased when walking speed is increased. The results agreed with the study of human locomotion using motion capture experiments.

From the clinical aspect of gait rehabilitation, functional recovery is maximized when locomotor training follows three major principles: practice, specificity and effort [31,32]. The first two principles emphasize that inducing correct joint kinematics to a patient in a repetitive manner results in better therapeutic outcomes and more practicing leads to improved motor learning. Therefore, it's important to plan correct motions for robotic trainers given that the patients will learn the gait pattern imposed to them during training process. It has been shown that variability in gait patterns is associated with differences in anatomical properties and walking speeds [19–23]. The present model is able to plan correct gait patterns in robotic gait rehabilitation by considering inter-subject difference and variability in walking speed. Thus, it can improve the therapeutic outcomes and help patients regain functional capacity.

A gait pattern prediction model and its applicability have been introduced. Some limitations and future directions will be discussed in next paragraphs to further improve the model. At a general level, the physical conditions of subjects with gait deficit can be divided into symmetric and asymmetric gait conditions. If the patients exhibit frozen joint or physical constraint that prevent the lower limb joints from achieving normal range of motion, they will belong to asymmetric condition. The joint kinematics of left and right legs are significantly different in this condition. The present model has not tailored to this group of patients in the current stage. Another limitation of this model is that it does not deal with asymmetries in anthropometric data of an individual.

GRNNs were implemented as non-linear mapping functions in the present model and the results are promising. However, neural network can only implicitly learn the distribution function of the observed data and the effect of each anthropometric factor on gait pattern has not been shown. An explicit regression function may further improve the present model.

The proposed model currently focuses only on lower limb joint kinematics. Planning pelvic motion also plays an important role in robotic gait rehabilitation [33]. We will examine whether the

proposed approach can be used in pelvic motion generation in the future research.

Acknowledgments

The authors would like to thank the Department of the Rehabilitation Medicine at the Tan Tock Seng Hospital (TTSH), the equipment and financial supports provided by the NMRC Research Grant 1051/2006 and the Robotics Research Center (RRC) of the Nanyang Technological University (NTU) for their support to this work. The authors would like to express their thanks to the reviewers for their constructive comments, which have greatly improved the quality of the manuscript. The authors would also like to thank Wei Qin Tean, Sze Jie Lim, Jun Jie Lim, and Thu Myint Aung for their assistance in the data collection.

Conflict of interest

The author does not have financial or personal relationship with other persons or organizations that might inappropriately influence the work presented therein.

References

- [1] Cai LL, Fong AJ, Otoshi CK, Liang Y, Burdick JW, Roy RR, et al. Implications of assist-as-needed robotic step training after a complete spinal cord injury on intrinsic strategies of motor learning. *J Neurosci* 2006;26(41):10564–68.
- [2] Riener R, Lünenburger L, Maier I, Colombo G, Dietz V. Locomotor training in subjects with sensori-motor deficits: an overview of the robotic gait orthosis lokomat. *J Healthc Eng* 2010;1(2):197–216.
- [3] Stauffer Y, Allemand Y, Bouri M, Fournier J, Clavel R, Metrailler P, et al. The WalkTrainer—a new generation of walking reeducation device combining orthoses and muscle stimulation. *IEEE Trans Neural Syst Rehabil Eng* 2009;17(1):38–45.
- [4] Schmidt H, Hesse S, Bernhardt R, Kruger J. HapticWalker—a novel haptic foot device. *ACM Trans Appl Percept* 2005;2(2):166–80.
- [5] Banala SK, Kim SH, Agrawal SK, Scholz JP. Robot assisted gait training with active leg exoskeleton (ALEX). *IEEE Trans Neural Syst Rehabil Eng* 2009;17(1):2–8.
- [6] Patton J, Brown DA, Peshkin M, Santos-Munn JJ, Makhlin A, Lewis E, et al. KineAssist. Design and development of a robotic overground gait and balance therapy device. *Top Stroke Rehabil* 2008;15(2):131–9.
- [7] Barbeau H. Locomotor training in neurorehabilitation: emerging rehabilitation concepts. *Neurorehabil Neural Repair* 2003;17(1):3–11.
- [8] Hasan MK, Park S-H, Seo S-J, Sohn D-H, Hwang S-H, Khang G. A gait rehabilitation and training system based on task specific repetitive approach. *IEEE ICBBE* 2009;1–4.
- [9] Winter DA. Biomechanics and motor control of human movement. John Wiley & Sons; 2004. p. 296–356.
- [10] Duschau-Wicke A, Von Zitzewitz J, Caprez A, Lunenburger L, Riener R. Path control: a method for patient-cooperative robot-aided gait rehabilitation. *IEEE Trans Neural Syst Rehabil Eng* 2010;18(1):38–48.
- [11] Colombo G, Joerg M, Schreiber R, Dietz V. Treadmill training of paraplegic patients using a robotic orthosis. *J Rehabil Res Dev* 2000;37(6):693–700.
- [12] Zoss AB, Kazerooni H, Chu A. Biomechanical design of the Berkeley lower extremity exoskeleton (BLEEX). *IEEE/ASME Trans Mechatron* 2006;11(2):128–38.
- [13] Schwartz MH, Rozumalski A, Trost JP. The effect of walking speed on the gait of typically developing children. *J Biomech* 2008;41(8):1639–50.
- [14] Lelas J. Predicting peak kinematic and kinetic parameters from gait speed. *Gait Posture* 2003;17(2):106–12.
- [15] Vaughan CL, Davis BL, O'Connor JC. Dynamics of human gait. Cape Town, South Africa: Kiboho Publishers; 1999. p. 25–7.
- [16] Chau T. A review of analytical techniques for gait data. Part 2. Neural network and wavelet methods. *Gait Posture* 2001;13(2):102–20.
- [17] Antonsson EK, Mann RW. The frequency content of gait. *J Biomech* 1985;18(1):39–47.
- [18] Beale MH, Hagan MT, Demuth HB. Neural network toolbox user's guide, vol. 8. The MathWorks Inc.; 2010. p. 12–4.
- [19] Kirtley C, Whittle MW, Jefferson RJ. Influence of walking speed on gait parameters. *J Biomed Eng* 1985;7(4):282–8.
- [20] Oberg T, Karsznia A, Oberg K. Joint angle parameters in gait: reference data for normal subjects, 10–79 years of age. *J Rehabil Res Dev* 1994;31(3):199–213.
- [21] Lelas JL, Merriman GJ, Riley PO, Kerrigan DC. Predicting peak kinematic and kinetic parameters from gait speed. *Gait Posture* 2003;17(2):106–12.
- [22] Hanlon M, Anderson R. Prediction methods to account for the effect of gait speed on lower limb angular kinematics. *Gait Posture* 2006;24(3):280–7.
- [23] Hof AL. Scaling gait data to body size. *Gait Posture* 1996;4(3):222–3.

- [24] Shuxia R, Dan Y, Fengqiu J, Xiushu T. Application of generalized regression neural network in prediction of cement properties. In: ICCDA June; 2010. V2-385–8.
- [25] Rustagi L, Kumar L, Pillai GN. Human gait recognition based on dynamic and static features using generalized regression neural network. In: ICMV; 2009. p. 64–8.
- [26] Findlow A, Goulermas JY, Nester C, Howard D, Kenney LPJ. Predicting lower limb joint kinematics using wearable motion sensors. *Gait Posture* 2008;28(1):120–6.
- [27] Kirtley C. Clinical gait analysis; 2011. <http://www.clinicalgaitanalysis.com>.
- [28] Luu TP, Low KH, Xingda Q, Lim HB, Hoon KH. On the development of an overground gait rehabilitation system: *NaTure-gaits*. In: IEEE HIC November; 2012.
- [29] Luu TP, Lim HB, Hoon KH, Xingda Q, Low KH. Subject-specific gait parameters prediction for robotic gait rehabilitation via generalized regression neural network. In: ROBIO; 2011.p. 914–9.
- [30] Perry J, Garrett M, Gronley JK, Mulroy SJ. Classification of walking handicap in the stroke population. *Stroke* 1995;26(6):982–9.
- [31] Werner C, Frankenberg SV, Treig T, Konrad M, Hesse S. Treadmill training with partial body weight support and an electromechanical gait trainer for restoration of gait in subacute stroke patients: a randomized crossover study. *Stroke* 2002;33:2895–901.
- [32] Ferris DP, Sawicki GS, Domingo AR. Powered lower limb orthoses for gait rehabilitation. *Top Spinal Cord Inj Rehabil* 2005;11:34–49.
- [33] Trueblood PR, Walker JM, Perry J, Gronley JK. Pelvic exercise and gait in hemiplegia. *Phys Ther* 1989;69(1):18–26.